Assignment I

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1) Satellite Orbiting Earth:

$$\ddot{r} = r\dot{\theta}^2 - \frac{K}{r^2} + u_1$$

$$\ddot{\theta} = \frac{-2\dot{r}\dot{\theta}}{r} + \frac{1}{r}u_2$$

r = Earth satellite distance measured from center

K = pos. constant.

θ = phase of the orbit

u_1, u_2 = control

a) $x_1 = r$

$x_2 = \dot{r}$

$$\dot{x}_1 = \dot{r} = x_2$$

$$\dot{x}_2 = \ddot{r} = x_1\dot{\theta}^2 - \frac{K}{x_1^2} + u_1$$

so $\boxed{\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1\dot{\theta}^2 - \frac{K}{x_1^2} + u_1\end{aligned}}$

$x_3 = \theta$

$x_4 = \dot{\theta}$

so $\dot{x}_3 = \dot{\theta} = x_4$

$\dot{x}_4 = \ddot{\theta} = \frac{-2\dot{r}\dot{\theta}}{r} + \frac{1}{r}u_2$

and $x_1 = r$ $x_2 = \dot{r}$

so

$$\boxed{\begin{aligned}x_3 &= x_4 \\ \dot{x}_4 &= \frac{-2x_2x_4}{x_1} + \frac{1}{x_1}u_2\end{aligned}}$$

b) $\ddot{r} = r\dot{\theta}^2 - \frac{K}{r^2}$ $\ddot{\theta} = \frac{-2\dot{r}\dot{\theta}}{r}$

$u_1 = 0$, $u_2 = 0$

$\boxed{x_1 = r,\ x_2 = \dot{r},\ x_3 = \theta,\ x_4 = \dot{\theta}}$

and $\dot{x}_1 = x_2$

$\dot{x}_2 = x_1 x_4^2 - \frac{K}{x_1^2} + u_1$

$\dot{x}_3 = x_4$

$\dot{x}_4 = \frac{-2x_2x_4}{x_1} + \frac{1}{x_1}u_2$

Equilibrium point satifies: $\dot{x} = f(x)$ where $f(x) = 0$

$\dot{x}_1 = \dot{x}_2 = \dot{x}_3 = \dot{x}_4 = 0$

$\dot{x}_1 = x_2 = 0$

$\boxed{x_2 = \dot{r} = 0}$ $\longrightarrow$ $\dot{x}_3 = x_4 = 0$

$\dot{x}_4 = \frac{-2x_2x_4}{x_1} + \underbrace{\frac{1}{x_1}u_2}_{=0}$

Since $x_1 = r > 0$ and $x_2 = 0$ and $x_4 = 0$

$\dot{x}_2 = \underbrace{\cancel{x_1x_4^2}}_{=0} - \frac{K}{x_1^2} \cancel{+u_1} = 0$

$-\frac{K}{x_1^2} = 0$

Since K is positive constant, the equation $-\frac{K}{x_1^2} = 0$ has no solution for any x_1 or r.

No equilibrium points exist.

Physical interpretation:

- if the satellite has no angular velocity $(\dot{\theta} = 0)$ and no input the earths gravity will always exert a force pulling the satellite towards it. ie. free fall towards earth.

c) $\ddot{r} = r\dot{\theta}^2 - \frac{K}{r^2} + u_1$

$\ddot{\theta} = -\frac{2\dot{r}\dot{\theta}}{r} + \frac{1}{r}u_2$

and $u_1 = \frac{K}{r^2}$ $u_2 = 0$

and $x_1 = r$, $x_2 = \dot{r}$, $x_3 = \theta$, $x_4 = \dot{\theta}$

$\dot{x}_1 = x_2$

$\dot{x}_2 = x_1x_4^2 - \frac{K}{x_1^2} + u_1$

$\dot{x}_3 = x_4$

$\dot{x}_4 = \frac{-2x_2x_4}{x_1} + \frac{1}{x_1}u_2$

Now we put in $u_1 = \frac{K}{x_1^2}$, $u_2 = 0$ $\longrightarrow$ $\dot{x}_1 = x_2 = 0$, $\dot{x}_3 = x_4 = 0$

$\dot{x}_2 = x_1x_4^2 - \cancel{\frac{K}{x_1^2}} + \cancel{\frac{K}{x_1^2}} = x_1x_4^2 = 0$

$\dot{x}_4 = -\frac{2\overset{0}{x_2}\overset{0}{x_4}}{x_1} + 0 = 0$

$\dot{x}_1 = x_2$

$\dot{x}_2 = x_1x_4^2$

$\dot{x}_3 = x_4$

$\dot{x}_4 = \frac{-2x_2x_4}{x_1}$

$\dot{x}_1 = \dot{x}_2 = \dot{x}_3 = \dot{x}_4 = 0$

$\dot{x}_1 = x_2 = 0$

$\therefore x_2 = 0$

$\dot{x}_2 = x_1x_4^2 = 0$

$$\therefore x_4 = 0 \rightarrow \dot{x}_3 = x_4 = 0$$

$$\dot{x}_4 = -\frac{2x_2 x_4}{x_1} = 0$$

$\therefore$ Equilibrium set:

$r > 0$, $\dot{r} = 0$, $\dot{\theta} = 0$, $\theta = \theta_e$ (any costant)

Physical interpretation:

The controller is giving an outward radial accel. equal to gravity.

d) $r(t) = p$, $\theta(t) = \omega t$, $u_1 = u_2 = 0$

$x_1 = r$, $x_2 = \dot{r}$, $x_3 = \theta$, $x_4 = \dot{\theta}$

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_1 x_4^2 - \frac{k}{x_1^2} + u_1$$
$$\dot{x}_3 = \dot{\theta} = x_4$$
$$\dot{x}_4 = \ddot{\theta} = -\frac{2x_2 x_4}{x_1} + \frac{1}{x_1} u_2$$

$\dot{x} = f(x, u)$

$x_1 = r$ and $r(t) = P \Rightarrow x_1 = P$

$x_2 = \dot{r}$ and $r(t) = P \Rightarrow x_2 = \dot{r} = \frac{d}{dt}(P) = 0 \Rightarrow x_2 = 0$

$x_3 = \theta$ and $\theta(t) = \omega t \Rightarrow x_3 = \omega t$

$x_4 = \dot{\theta}$ and $\theta(t) = \omega t \Rightarrow x_4 = \dot{\theta} = \frac{d}{dt}(\omega t) = \omega \Rightarrow x_4 = \omega$

$$x(t) = \begin{bmatrix} P \\ 0 \\ \omega t \\ \omega \end{bmatrix} \qquad u(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\delta x = x - x_{Ref}$$

$\therefore$ $\delta r = r - p$ $\quad$ $\delta \dot{r} = \dot{r} - 0$

$\delta\theta = \theta - \omega t$ $\quad$ $\delta\dot{\theta} = \dot{\theta} - \omega$

$$\dot{x}_2 = x_1 x_4^2 - \frac{k}{x_1^2} + u_1$$

$$0 = P\omega^2 - \frac{k}{P^2} + 0$$

$$\frac{k}{P^2} = P\omega^2 \Rightarrow \boxed{k = P^3\omega^2}$$

Now jacobian matrix:

$$\dot{x} = f(x, u)$$

$$\delta\dot{x} = A(t)\,\delta x + B(t)\,\delta u$$

$\frac{\partial f}{\partial u} \Rightarrow B_{ij} = \frac{\partial f_i}{\partial u_j}$

$\frac{\partial f}{\partial x} \Rightarrow A_{ij} = \frac{\partial f_i}{\partial x_j}$

$$f_1(t) = x_2$$
$$f_2(t) = x_1 x_4^2 - \frac{k}{x_1^2} + u_1$$
$$f_3(t) = x_4$$
$$f_4(t) = \frac{-2x_2 x_4}{x_1} + \frac{1}{x_1} u_2$$

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \cdots & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \cdots & \frac{\partial f_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ x_4^2 + \frac{2k}{x_1^3} & 0 & 0 & 2x_1 x_4 \\ 0 & 0 & 0 & 1 \\ & -\frac{2x_4}{x_1} & 0 & -\frac{2x_2}{x_1} \end{bmatrix}$$

$$\frac{2x_2 x_4}{x_1^2} - \frac{u_2}{x_1^2}$$

Similarly:

$$B = \frac{\partial f}{\partial u} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_1} \\ \frac{\partial f_4}{\partial u_1} & \frac{\partial f_4}{\partial u_1} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & \frac{1}{x_1} \end{bmatrix}$$

Now, $x_1 = P$, $x_2 = 0$, $x_3 = \omega t$, $x_4 = \omega$, $k = P^3\omega^2$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega^2 & 0 & 0 & 2P\omega \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2\omega}{P} & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & \frac{1}{P} \end{bmatrix}$$

$$\boxed{\delta\dot{x} = A\,\delta x + B\,\delta u}$$

2a) $\dot{x} = v\cos\theta$ $\qquad$ $z = (x, y)$

$\dot{y} = v\sin\theta$

$\dot{\theta} = \omega$

$\dot{v} = a$

$x = z_1$, $\quad y = z_2$

$\dot{z} = (\dot{x}, \dot{y})$

$\dot{x} = v\cos\theta \Rightarrow \dot{x}^2 = v^2\cos^2\theta$

$\dot{y} = v\sin\theta \Rightarrow \dot{y}^2 = v^2\sin^2\theta$

$$\dot{x}^2 + \dot{y}^2 = v^2\cos^2\theta + v^2\sin^2\theta$$
$$= v^2 \underbrace{(\cos^2\theta + \sin^2\theta)}_{=1}$$

$$\dot{x}^2 + \dot{y}^2 = v^2$$
$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$\dot{x} = v\cos\theta \qquad \dot{y} = v\sin\theta$

$$\frac{\dot{y}}{\dot{x}} = \frac{\cancel{v}\sin\theta}{\cancel{v}\cos\theta}$$

$= \sin\theta$ $\Big\}$ $\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cos\theta$

$$\frac{\dot{y}}{\dot{x}} = \tan\theta$$

$$\theta = \arctan\left(\frac{\dot{y}}{\dot{x}}\right) = \operatorname{arctan2}(\dot{y}, \dot{x})$$

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$a = \dot{v} = \frac{\dot{x}\ddot{x} + \dot{y}\ddot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$$

$$\theta = \operatorname{arctan2}(\dot{y}, \dot{x})$$

$$w = \dot{\theta} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2 + \dot{y}^2}$$

$$= \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{v^2}$$

$$(x, y, \theta, v) = \beta(z, \dot{z}) = \left(z_1, z_2, \operatorname{atan2}(\dot{z}_2, \dot{z}_1), \sqrt{\dot{z}_1^2 + \dot{z}_2^2}\right)$$

$$(w, a) = \gamma(z, \dot{z}, \ddot{z}) = \left(\frac{\dot{z}_1\ddot{z}_2 - \dot{z}_2\ddot{z}_1}{\dot{z}_1^2 + \dot{z}_2^2}, \frac{\dot{z}_1\ddot{z}_1 + \dot{z}_2\ddot{z}_2}{\sqrt{\dot{z}_1^2 + \dot{z}_2^2}}\right)$$

b) IC: $t=0$
FC: $t=T=10$

$$\begin{bmatrix} x(0) \\ y(0) \\ \theta(0) \\ v(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} x(T) \\ y(T) \\ \theta(T) \\ v(T) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{\pi}{2} \\ 1 \end{bmatrix}$$

$$x(t) = \sum_{i=0}^{3} b_{0i}\,\psi_i(t) = b_{00}(1) + b_{01}(t) + b_{02}(t^2) + b_{03}(t^3)$$

$$y(t) = \sum_{i=0}^{3} b_{1i}\,\psi_i(t) = b_{10}(1) + b_{11}(t) + b_{12}(t^2) + b_{13}(t^3)$$

$$\dot{x}(t) = b_{01} + 2b_{02}t + 3b_{03}t^2$$
$$\dot{y}(t) = b_{11} + 2b_{12}t + 3b_{13}t^2$$

At $t=0$:

$x(0)=0$, $y(0)=0$, $\theta(0)=0$, $v(0)=1$

Since $\theta(0)=0$, the velocity points along $+x$ axis

So: $\dot{y}(0)=0$
$\dot{x}(0) > 1$

The $v(0)=1$

$$v(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$v(0) = \sqrt{\dot{x}(0)^2 + \underbrace{\dot{y}(0)^2}_{=0}}$$

$$= \sqrt{\dot{x}(0)^2}$$

$$1 = \sqrt{\dot{x}(0)^2}$$

$$\dot{x}(0) = \pm 1 \rightarrow \dot{x}(0) = 1$$

So at $t=0$: $x(0)=0$, $y(0)=0$, $\dot{x}(0)=1$, $\dot{y}(0)=0$

For $t=T=10$: $x(10)=0$, $y(10)=0$, $\theta(10)=\frac{\pi}{2}$, $v(\theta)=1$

Because $\theta(10)=\frac{\pi}{2}$, means velocity points in $+y$ direction

So: $\dot{x}(10)=0$, $\dot{y}(10)>0$

and $v(10)=1$

$$v(10) = \sqrt{\dot{x}(10)^2 + \dot{y}(10)^2}$$

$$1 = \sqrt{0 + \dot{y}(10)^2}$$

$$\dot{y}(10) = 1$$

So: $x(10)=0$, $y(10)=0$, $\dot{x}(10)=0$, $\dot{y}(10)=1$

Now plug $t=0$ into the sums above:

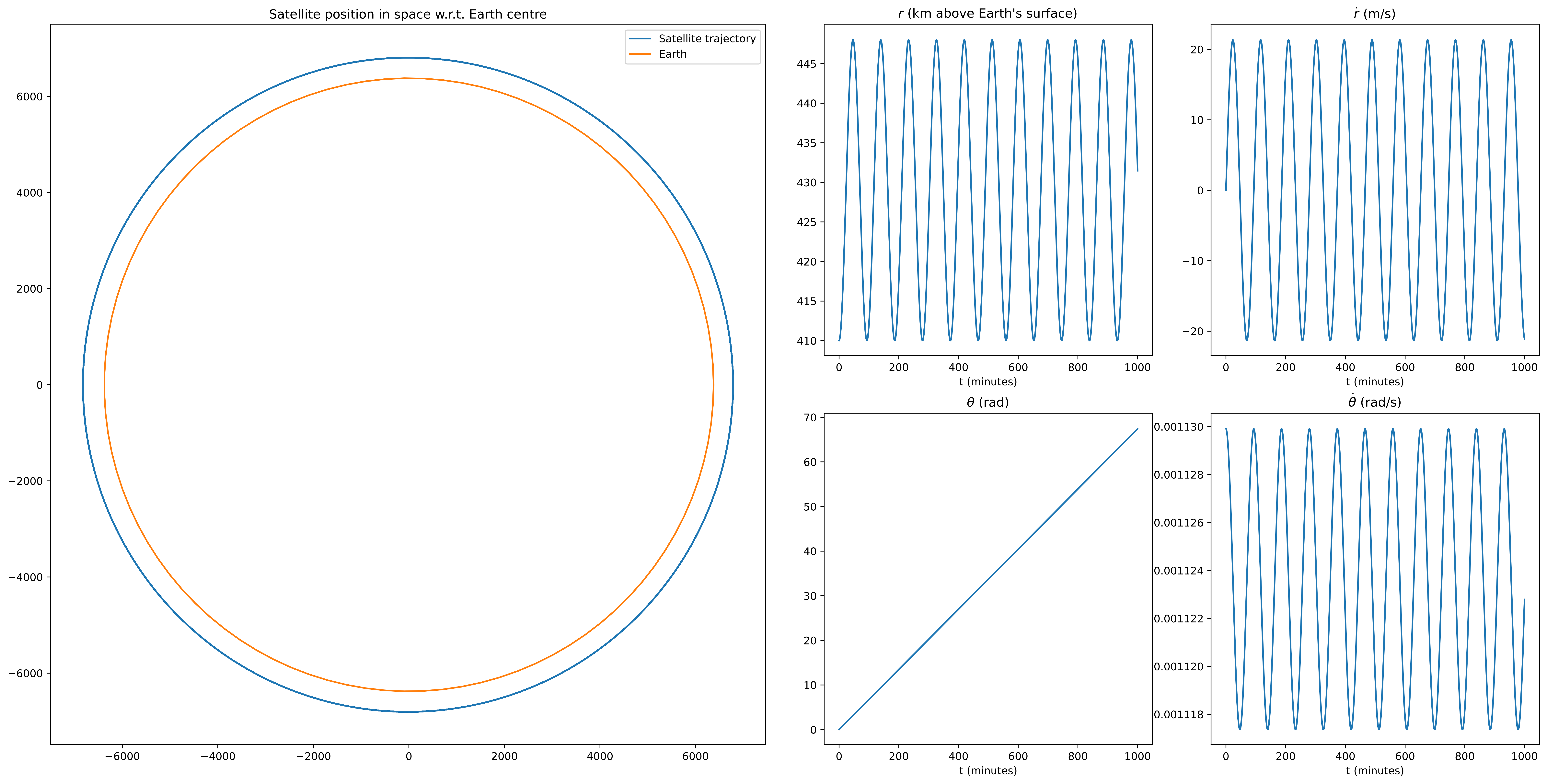
$$x(0) = b_{00} = 0$$
$$y(0) = b_{10} = 0$$
$$\dot{x}(0) = b_{01} = 1$$
$$\dot{y}(0) = b_{11} = 0$$

and at $t=10$

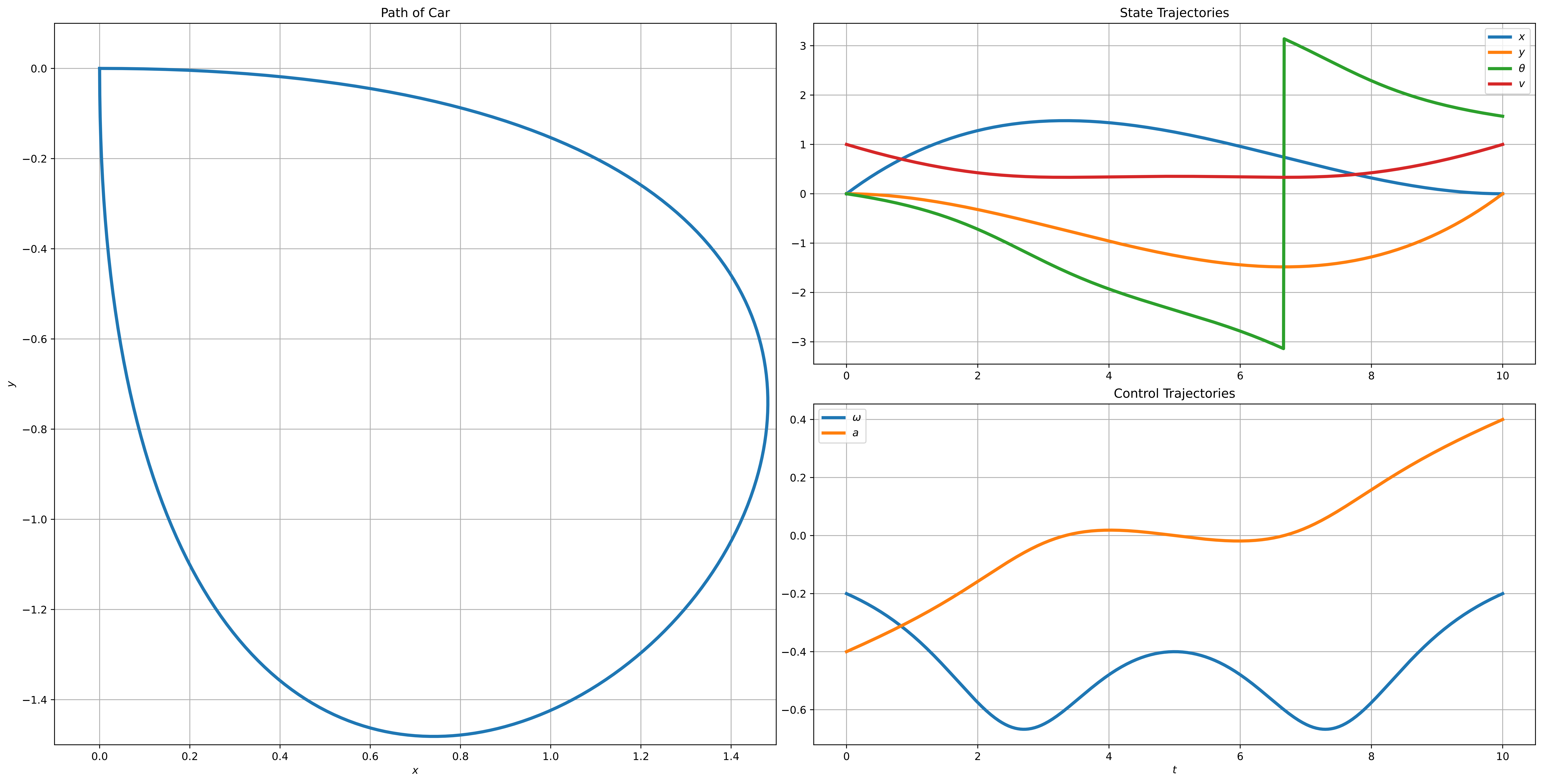
$$x(10) = b_{00} + 10b_{01} + 100b_{02} + 1000b_{03} = 0$$
$$y(10) = b_{10} + 10b_{11} + 100b_{12} + 1000b_{13} = 0$$
$$\dot{x}(10) = b_{01} + 20b_{02} + 300b_{03} = 0$$
$$\dot{y}(10) = b_{11} + 20b_{12} + 300b_{13} = 1$$

$\therefore$

$$\boxed{\begin{aligned} b_{00} &= 0 \\ b_{10} &= 0 \\ b_{01} &= 1 \\ b_{11} &= 0 \\ b_{00} + 10b_{01} + 100b_{02} + 1000b_{03} &= 0 \\ b_{10} + 10b_{11} + 100b_{12} + 1000b_{13} &= 0 \\ b_{01} + 20b_{02} + 300b_{03} &= 0 \\ b_{11} + 20b_{12} + 300b_{13} &= 1 \end{aligned}}$$



Satellite position in space w.r.t. Earth centre
Satellite trajectory
Earth
r (km above Earth's surface)
$\dot{r}$ (m/s)
θ (rad)
$\dot{\theta}$ (rad/s)
t (minutes)


Path of Car
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
x
y
State Trajectories
x
y
θ
v
3
2
1
0
−1
−2
−3
0
2
4
6
8
10
Control Trajectories
ω
a
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
2
4
6
8
10
t